Professional Practices (HUM-305)

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Chap 2: PRIVACY

- Privacy Risks and Principles
- What Is Privacy?
- New Technology, New Risks
- Terminology and Principles for Managing Personal Data
- New Technologies, Supreme Court Decisions, and Expectation of Privacy
- Search and Seizure of Computers and Phones Video Surveillance and Face Recognition
- Marketing and Personalization
- Our Social and Personal Activity
- Location Tracking

Privacy

What is Privacy?

- **Privacy** means the right of individuals to control how their personal information is collected, used, stored, and shared.

In IT, privacy mainly concerns:

- Personal data
- Digital identity
- Online behavior
- Communication records

Privacy

Privacy Risks

Definition:

- Privacy risks are potential dangers that may lead to unauthorized access, misuse, exposure, or loss of personal information.

Major Privacy Risks (with CS/IT Examples)

1: Data Breaches

- When hackers gain unauthorized access to databases.
- 👉 Example: A cyberattack exposing user passwords from a company like Facebook.

2: Identity Theft

- When someone steals personal information to impersonate another person.

Example:
Using stolen CNIC or credit card details for fraud.

Privacy

3: Unauthorized Data Sharing

- Organizations sharing user data without consent.

Example:
An app selling user location data to third parties.

4: Surveillance & Tracking

- Continuous monitoring of user activity online.

Example:
Tracking browsing behavior through cookies by platforms like Google.

5: Insider Threats

- Employees misusing customer data.

Example:
A database administrator copying customer phone numbers.

Privacy

6: Weak Security Controls

- Poor encryption, weak passwords, or outdated systems.

Example:
Storing passwords in plain text instead of encrypted form.

7: AI & Big Data Misuse

- Artificial Intelligence analyzing personal data without transparency.

Example:
Facial recognition systems collecting biometric data without consent.

Privacy Principles

- Privacy principles are guidelines that organizations must follow to protect personal information.

Many global frameworks define these principles, such as:

- OECD Privacy Guidelines
- General Data Protection Regulation (GDPR)

Privacy Principles

Core Privacy Principles

1: Lawfulness, Fairness & Transparency

- Data must be collected legally and transparently.

Users must know:

- What data is collected
- Why it is collected
- How it will be used

2: Purpose Limitation

- Data should be collected for a specific purpose only.

Example:
If data is collected for account creation, it should not be used for marketing without consent.

Privacy Principles

3: Data Minimization

- Collect only necessary data.

Example:
An online form should not ask for marital status if not required.

4: Accuracy

- Data must be correct and updated.

Example:
Allow users to update their profile information.

5: Storage Limitation

- Do not store data longer than necessary.

Example:
Delete inactive user accounts after a specific period.

Privacy Principles

6: Integrity & Confidentiality (Security)

Protect data using:

- Encryption
- Firewalls
- Access controls
- Authentication

7: Accountability

- Organizations must be responsible for protecting user data.

Example:
Appointing a Data Protection Officer (DPO).

Types of Privacy

Types of Privacy

- 1 **Information Privacy** – Protection of personal data
- 2 **Communication Privacy** – Protection of emails, calls, messages
- 3 **Physical Privacy** – Freedom from surveillance
- 4 **Decisional Privacy** – Freedom to make personal choices

New Technology, New Risks

Modern technologies bring convenience but also increase privacy risks.

Examples of New Technologies & Risks

1: Artificial Intelligence (AI)

- Collects and analyzes massive personal data.
- Risk: Biased decisions, facial recognition misuse.

2: Social Media Platforms

- Like Facebook and Instagram.
- Risks:
- Data mining
- Targeted advertising
- Identity theft

New Technology, New Risks

3: Cloud Computing

- Data stored on remote servers.

Risk:

- Data breaches
- Unauthorized access

4: Internet of Things (IoT)

- Smart devices (smart watches, CCTV, smart homes).

Risk:

- Continuous surveillance
- Location tracking

New Technology, New Risks

5: Big Data & Data Analytics

- Companies like Google collect browsing history.

Risk:

- Profiling users without consent
- Behavioral tracking

Why New Risks Occur?

- Large-scale data collection
- Weak cybersecurity
- Lack of user awareness
- Complex AI algorithms

Terminology and Principles for Managing Personal Data

Important Terminology

1: Personal Data

- Any information that identifies a person. Example: Name, CNIC, email, IP address.

2: Data Subject

- The individual whose data is collected.

3: Data Controller

- Organization that decides how data is used.

Terminology and Principles for Managing Personal Data

4: Data Processor

- Entity that processes data on behalf of the controller.

5: Sensitive Data

Highly confidential data:

- Health records
- Biometric data
- Financial information

Principles for Managing Personal Data

Many laws like the General Data Protection Regulation (GDPR) define these principles:

1 Lawfulness & Transparency

- Collect data legally and inform users.

2 Purpose Limitation

- Use data only for specific purposes.

3 Data Minimization

- Collect only necessary data.

Principles for Managing Personal Data

4 Accuracy

- Keep data correct and updated.

5 Storage Limitation

- Do not store data forever.

6 Security (Integrity & Confidentiality)

- Use encryption and strong access control.

7 Accountability

- Organizations must be responsible for data protection.

New Technologies, Supreme Court Decisions, and Expectation of Privacy

Expectation of Privacy

- It means what level of privacy a person reasonably expects in a situation.

Example:

- You expect privacy in your home.
- You expect less privacy on public roads.

New Technologies, Supreme Court Decisions, and Expectation of Privacy

Role of Supreme Court

- Courts decide how privacy rights apply to new technologies.

Example 1:

In the United States, the Supreme Court of the United States has ruled that:

- Police need a warrant to access mobile phone data in certain cases.
- GPS tracking without permission can violate privacy rights.
- These decisions show that even with new technologies, privacy rights must be protected.

Technology vs Law Challenge

- Technology evolves fast.
- Law develops slowly.
- Courts interpret constitutional rights in modern contexts.

CS / IT Professional Responsibility

CS / IT Professional Responsibility

As an IT professional, you must:

- Design privacy-by-design systems.
- Encrypt databases.
- Protect user credentials.
- Avoid unauthorized surveillance.
- Follow ethical standards like those from Association for Computing Machinery

Topic	Key Idea
Privacy	Right to control personal information
New Technology	Creates new privacy risks
Terminology	Defines roles and data types
Principles	Rules to protect personal data
Supreme Court	Interprets privacy in modern tech context

Search and Seizure of Computers and Phones

1) Search and Seizure of Computers and Phones

Meaning

- Search and seizure refers to the legal authority of law enforcement agencies to search digital devices (computers, laptops, smartphones, servers) and confiscate them as evidence in criminal investigations.

Why It Is Important in IT

Computers and phones store:

- Emails
- Messages (WhatsApp, Messenger)
- Photos and videos
- Browsing history
- Financial records
- Cloud data
- So digital devices often contain critical evidence.

Search and Seizure of Computers and Phones

Legal Framework

- In the United States, digital searches are governed under the **Fourth Amendment**, which protects against unreasonable searches and seizures.

A landmark case:

- Riley v. California
  The Supreme Court ruled that police must obtain a **warrant** before searching a mobile phone seized during an arrest.

Issues in Digital Search

- Access to encrypted data
- Cloud storage jurisdiction problems
- Scope of warrant (how much data can be searched?)
- Privacy of unrelated third parties

Ethical Concerns

- Mass data copying
- Retaining irrelevant personal data
- Violating user privacy rights

Video Surveillance and Face Recognition

2) Video Surveillance and Face Recognition

Video Surveillance

- Use of cameras (CCTV, IP cameras, smart cameras) to monitor public and private spaces.

Examples:

- Airports
- Universities
- Shopping malls
- Smart cities

Benefits:

- Crime prevention
- Evidence collection
- Public safety

Risks:

- Constant monitoring
- Chilling effect on freedom
- Misuse by authorities

Face Recognition Technology (FRT)

Face Recognition Technology (FRT)

- Face recognition uses AI algorithms to identify individuals from images or video.

How It Works:

- Capture face image
- Extract facial features
- Match with database

Real-world Example:

- Clearview AI – Scraped billions of images from the internet to create a face database.

Privacy Concerns:

- Biometric data misuse
- False positives (especially minorities)
- No consent from individuals
- Mass surveillance risks

Marketing and Personalization

3) Marketing and Personalization

What Is It?

- Companies collect user data to show personalized ads, recommendations, and content.

Example:

- Google uses browsing history to personalize ads.
- Facebook tracks likes, shares, and interests to target advertisements.

Data Used:

- Search history, Purchase history, Location data, Social interactions, Device information

Benefits:

- Relevant ads, Better user experience, Customized content

Risks:

- Manipulation of behavior, Political micro-targeting
- Data breaches, Loss of autonomy

Our Social and Personal Activity (Digital Footprint)

4) Our Social and Personal Activity (Digital Footprint)

What Is Digital Footprint?

- The trail of data we leave online.

Active Footprint:

- Posting on social media, Commenting, Uploading photos

Passive Footprint:

- Cookies, Background tracking, IP logging

Platforms Collecting Data:

- Instagram, TikTok, LinkedIn

Risks:

- Identity theft, Reputation damage, Profiling, Cyberstalking

Location Tracking

5) Location Tracking

What Is Location Tracking?

Monitoring and recording a person's geographical position via:

- GPS, Wi-Fi, Cell towers, Bluetooth beacons

Example:

- Apple and Google collect location history for services like maps and recommendations.

Legal Case:

- Carpenter v. United States
  The court ruled that accessing historical cell phone location records requires a warrant.

Risks:

- Stalking, Government mass tracking, Behavioral profiling, Loss of anonymity

Conclusion

Conclusion

For Computer Science and IT professionals:

- Understand privacy laws
- Implement data minimization
- Use encryption
- Follow ethical data practices
- Apply Privacy by Design principles

- Technology creates convenience, but it also introduces serious privacy and ethical challenges. IT professionals must balance **innovation with responsibility**